WebClient Computing

CSS Basics

Department of Computer Science

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Announcements

- Exam date
  - 22nd (Friday)
- Covers
  - HTML
  - CSS
- Project
  - Opportunity to apply what you learned from the course
  - 20% of total credit
  - Interview with an instructor one week after the exam
- Survey review

Horizontal Navigation Bar

- By default, <li> tag is a block element and takes the entire width
- Have <li> tag to inline element makes it align in a same line

Home News Contact About

```html
<!DOCTYPE html>
<html>
<head>
<style>
  ul {
    list-style-type: none;
    margin: 0;
    padding: 0;
  }

  li {
    display: inline;
  }
</style>
</head>
<body>

<ul>
  <li><a href="#home">Home</a></li>
  <li><a href="#news">News</a></li>
  <li><a href="#contact">Contact</a></li>
  <li><a href="#about">About</a></li>
</ul>

</body>
</html>
```

Make It Better

Home News Contact About

- li - display: inline
  - Have it in horizontal view
- li a - display: inline-block
  - Remember the difference between block and inline elements
  - Have an effect of block in an line
  - Change it to “inline” or ”block”

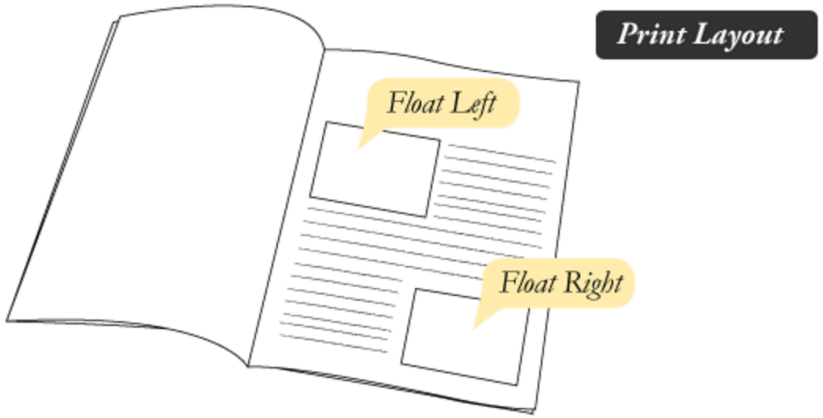
```
<style>
  ul {
    list-style-type: none;
    margin: 0;
    padding: 0;
    background-color: #333;
  }
  li {
    display: inline;
    font-family: arial, sans;
  }
  li a {
    display: inline-block;
    color: white;
    text-align: center;
    padding: 14px 16px;
    text-decoration: none;
  }

  li a:hover {
    background-color: #111;
  }
</style>
```

Can You Make This?

Home News Contact About

- Tips
  - make a cell goes moves around - float



https://css-tricks.com/all-about-floats/

Locate a Cell Right

```
.right-align {
  float: right;
}
```

```
<ul>
  <li><a href="#home">Home</a></li>
  <li><a href="#news">News</a></li>
  <li><a href="#contact">Contact</a></li>
  <li class="right-align"><a href="#about">About</a></li>
</ul>
```

Home News Contact About

Navigating With the Bar

Home News Contact About

This is H1 heading

This is H2 heading

This is H3 heading

This is H4 heading

This is H5 heading

This is H6 heading

This is H1 heading

This is H1 heading

This is H2 heading

This is H3 heading

This is H4 heading

This is H5 heading

This is H6 heading

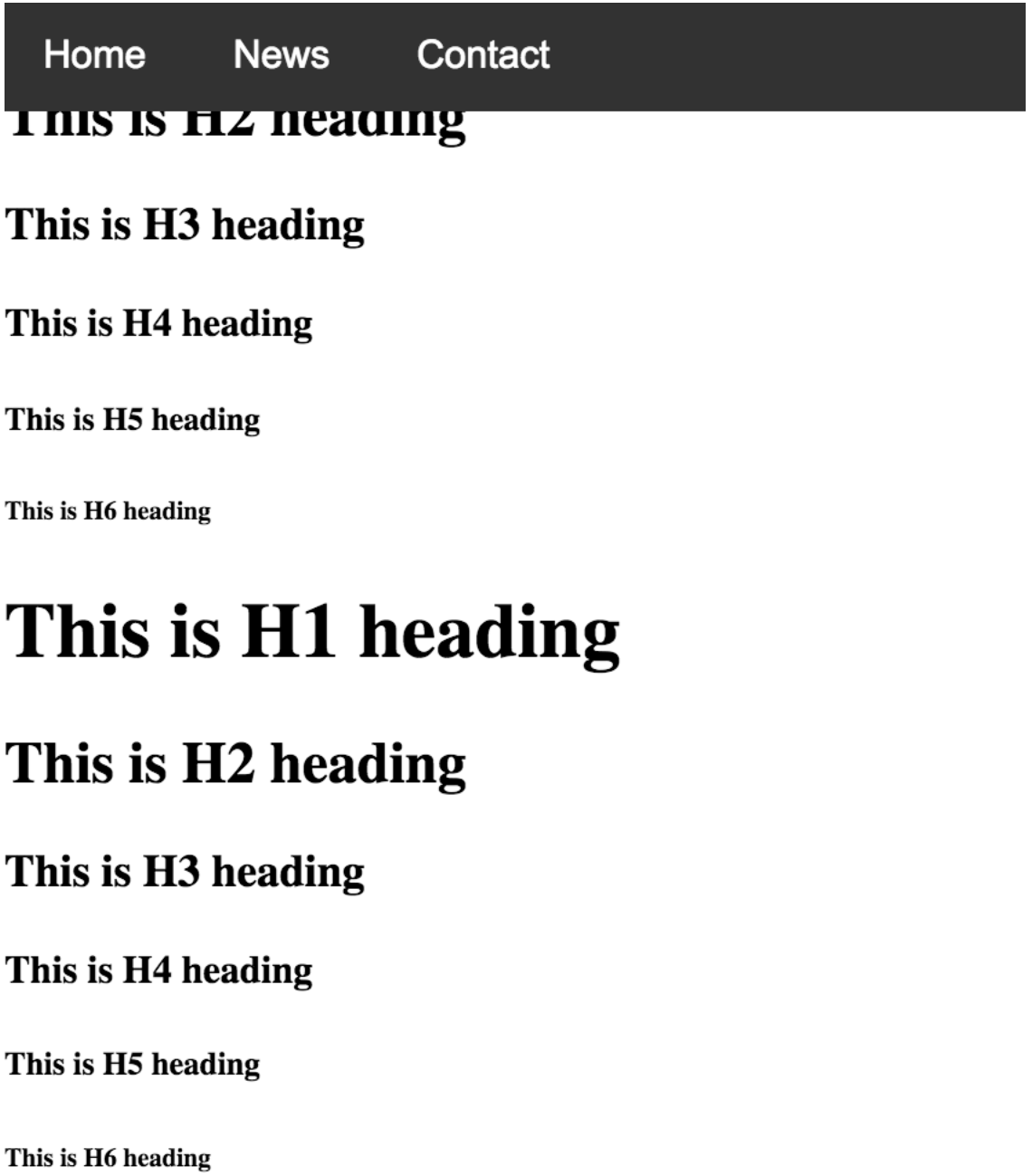
This is H1 heading

This is H2 heading

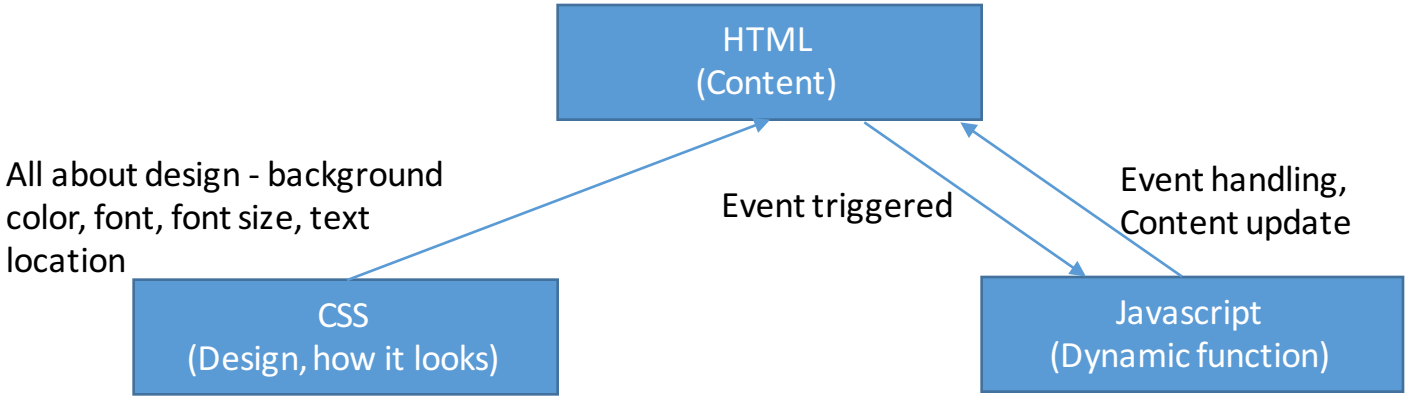
Fixed Horizontal Navigation Bar

```
ul {
  list-style-type: none;
  margin: 0;
  padding: 0;
  background-color: #333;
  position: fixed;
  top: 0;
  width: 100%;
}
```

- top, left, right, bottom sets fixed position
- margin sets outer spaces
- Fixed makes elements to align with other elements



HTML, CSS, and Javascript



- HTML – Hyper Text Markup Language
  - House itself
- CSS - Painting the house and decoration
- Javascript – rich features, electricity on/off using a smartphone

What is Javascript

- Javascript is an interpreted programming language with OO capabilities
- Most commonly used in web browsers
- Design goals
  - Enable interaction with users
  - Easily control web browser
  - Alter document in a web browser window
- Javascript language is well standardized and compatibility among browsers is good

Myths about Javascript

- Javascripts has something to do with Java
  - No. The name is marketing purpose
  - Script language for Java – Scala
- Javascript is for non-programmers
  - Though it is an interpreted script language, it is a full-featured programming language

Javascript Demo

- Have a popup window!

```
alert("Hello Javascript");
```

helloworld.js

```
<!DOCTYPE html>
<html>
<head>
  <title>Javascript exercise</title>
  <script type="text/javascript" src="helloworld.js"></script>
</head>
<body>
</body>
</html>
```

index.html

Javascript Variable

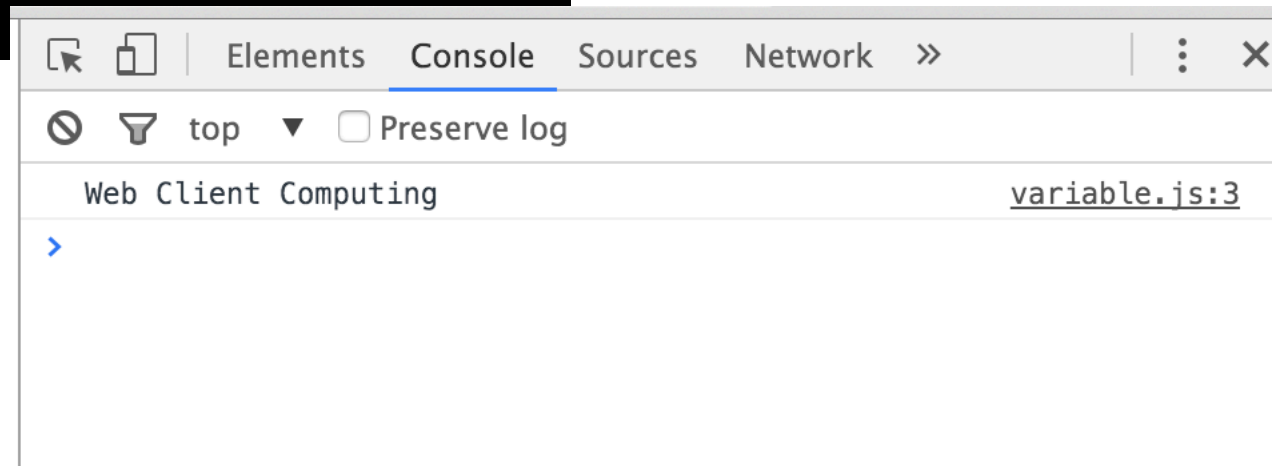
```
var class_name = "Web Client Computing";
alert(class_name);
console.log(class_name);
```

variable.js

```
<!DOCTYPE html>
<html>
<head>
  <title>Javascript exercise</title>
  <script type="text/javascript" src="variable.js"></script>
</head>
<body>
</body>
</html>
```

index.html

Chrome developer tools



Javascript Getting Input

```javascript
var name = prompt("What is your name: ");
console.log("name is ", name);
```

input.js

```html
<!DOCTYPE html>
<html>
<head>
  <title>Javascript exercise</title>
  <script type="text/javascript" src="input.js"></script>
</head>
<body>
</body>
</html>
```

index.html

Javascript Elementary Rules

- Comments for are same as C, C++, Java
  - //
  - /* ….. */
- Literals – data values in a program

```
12              // The number twelve
1.2             // The number one point two
"hello world"   // A string of text
'Hi'            // Another string
true            // A Boolean value
false           // The other Boolean value
/javascript/gi  // A "regular expression" literal
null            // Absence of an object
```

Numbers in Javascript

- All numbers in 64 bit floating point
- Recognizes
  - Base-10 integer: 3421
  - Base-16 begins with 0X: 0XCAFE911
- Floating point literals
  - 3.14
  - .33333
  - 6.02e23 //$6.02*10^{23}$

Exercises

```
> var a=100;
<· undefined
> var b = 3.141592;
<· undefined
> var c = 3.2e2;
<· undefined
```

```
> typeof(a)
<· "number"
> typeof(b)
<· "number"
> typeof(c)
<· "number"
```

Reference

- W3 Schools CSS